

Medical Imaging Dataset Analysis Report

=====

Dataset: Task09_Spleen
Generated: 2026-06-30 09:46
Task: segmentation (dataset.json contains image/label training pairs.)

Dataset Overview

=====

Dataset root: C:\Users\Ielen\Desktop\Recherches stage\Stage
Japon\Hokkaido\Stage\check_data_project\check_data\dataset\Task09_Spleen
Split: all
Patients detected: 61
Patients analysed: 61
Failed analyses: 0
Patients with missing data: 20
Total storage: 1.61 GB
Minimum memory: 176.16 MB

Task Detection

=====

Detected task: segmentation
Reason: dataset.json contains image/label training pairs.
Labels found: Yes
Label files detected: 41
Image/label pairs: 41

Missing Data

=====

Patient ID	Split	Missing data

spleen_1	test	annotation_label
spleen_11	test	annotation_label
spleen_15	test	annotation_label
spleen_23	test	annotation_label
spleen_30	test	annotation_label
spleen_34	test	annotation_label
spleen_35	test	annotation_label
spleen_36	test	annotation_label
spleen_37	test	annotation_label
spleen_39	test	annotation_label
spleen_42	test	annotation_label
spleen_43	test	annotation_label
spleen_48	test	annotation_label
spleen_50	test	annotation_label
spleen_51	test	annotation_label
spleen_54	test	annotation_label
spleen_55	test	annotation_label
spleen_57	test	annotation_label
spleen_58	test	annotation_label
spleen_7	test	annotation_label

Memory

=====

Minimum memory needed: 176.16 MB
Basis: Largest image loaded as float32, which is how analyze_nifti_volume reads CT voxel data.
Largest native volume: 176.16 MB
Largest float32 volume: 176.16 MB

Slices

=====

Dimensionality | Patients | %

----- | ----- | ---
3D | 61 | 100

Slice count | Patients | %

-----		-----		----
94		3		4.92
103		2		3.28
112		2		3.28
168		2		3.28
44		2		3.28
48		2		3.28
50		2		3.28
61		2		3.28
89		2		3.28
96		2		3.28
98		2		3.28
100		1		1.64
101		1		1.64
102		1		1.64
104		1		1.64
114		1		1.64
119		1		1.64
129		1		1.64
135		1		1.64
148		1		1.64
156		1		1.64
157		1		1.64
164		1		1.64
31		1		1.64
33		1		1.64
34		1		1.64
36		1		1.64
38		1		1.64
40		1		1.64
41		1		1.64
51		1		1.64
54		1		1.64
55		1		1.64
56		1		1.64
60		1		1.64
65		1		1.64
67		1		1.64
77		1		1.64
80		1		1.64
83		1		1.64
86		1		1.64
87		1		1.64
88		1		1.64
90		1		1.64
92		1		1.64
93		1		1.64
95		1		1.64
97		1		1.64
99		1		1.64

Thickness (mm) | Patients | %

-----		-----		----
5		41		67.21
2.5		6		9.84
1.5		4		6.56
4		4		6.56
1.6		2		3.28
3		1		1.64
7		1		1.64
7.5		1		1.64
8		1		1.64

Voxel Size and Volume

=====

Voxel size XxYxZ (mm) | Patients | %

-----		-----		----
0.977 x 0.977 x 5		3		4.92
0.684 x 0.684 x 5		2		3.28
0.703 x 0.703 x 5		2		3.28
0.74 x 0.74 x 5		2		3.28
0.797 x 0.797 x 5		2		3.28
0.613 x 0.613 x 2.5		1		1.64
0.633 x 0.633 x 5		1		1.64
0.643 x 0.643 x 5		1		1.64
0.647 x 0.647 x 5		1		1.64
0.66 x 0.66 x 5		1		1.64
0.676 x 0.676 x 5		1		1.64
0.699 x 0.699 x 7		1		1.64
0.719 x 0.719 x 2.5		1		1.64
0.727 x 0.727 x 2.5		1		1.64
0.734 x 0.734 x 5		1		1.64
0.738 x 0.738 x 5		1		1.64
0.741 x 0.741 x 5		1		1.64
0.742 x 0.742 x 2.5		1		1.64
0.742 x 0.742 x 5		1		1.64
0.744 x 0.744 x 5		1		1.64
0.75 x 0.75 x 5		1		1.64
0.752 x 0.752 x 5		1		1.64
0.754 x 0.754 x 1.5		1		1.64
0.77 x 0.77 x 4		1		1.64
0.773 x 0.773 x 5		1		1.64
0.785 x 0.785 x 5		1		1.64
0.785 x 0.785 x 7.5		1		1.64
0.793 x 0.793 x 5		1		1.64
0.793 x 0.793 x 8		1		1.64
0.795 x 0.795 x 1.6		1		1.64
0.795 x 0.795 x 5		1		1.64
0.803 x 0.803 x 5		1		1.64
0.805 x 0.805 x 4		1		1.64
0.828 x 0.828 x 5		1		1.64
0.836 x 0.836 x 3		1		1.64
0.838 x 0.838 x 5		1		1.64
0.852 x 0.852 x 5		1		1.64
0.854 x 0.854 x 4		1		1.64
0.859 x 0.859 x 2.5		1		1.64
0.859 x 0.859 x 5		1		1.64
0.871 x 0.871 x 5		1		1.64
0.889 x 0.889 x 5		1		1.64
0.902 x 0.902 x 5		1		1.64

0.904 x 0.904 x 5	1	1.64
0.906 x 0.906 x 5	1	1.64
0.92 x 0.92 x 1.6	1	1.64
0.924 x 0.924 x 5	1	1.64
0.93 x 0.93 x 5	1	1.64
0.934 x 0.934 x 1.5	1	1.64
0.938 x 0.938 x 5	1	1.64
0.951 x 0.951 x 5	1	1.64
0.961 x 0.961 x 4	1	1.64
0.967 x 0.967 x 1.5	1	1.64
0.969 x 0.969 x 1.5	1	1.64
0.977 x 0.977 x 2.5	1	1.64
Physical volume XxYxZ (mm) Patients %		
-----	-----	----
360 x 360 x 480	2	3.28
314 x 314 x 237.5	1	1.64
324 x 324 x 495	1	1.64
329 x 329 x 170	1	1.64
331.264 x 331.264 x 240	1	1.64
337.92 x 337.92 x 430	1	1.64
346 x 346 x 400	1	1.64
350 x 350 x 220	1	1.64
350 x 350 x 325	1	1.64
358 x 358 x 230.999	1	1.64
368 x 368 x 217.5	1	1.64
372 x 372 x 222.5	1	1.64
376 x 376 x 470	1	1.64
378 x 378 x 200	1	1.64
379 x 379 x 190	1	1.64
379 x 379 x 280	1	1.64
379.392 x 379.392 x 490	1	1.64
380 x 380 x 192.5	1	1.64
380 x 380 x 445	1	1.64
381 x 381 x 205	1	1.64
384 x 384 x 465	1	1.64
385 x 385 x 300	1	1.64
386 x 386 x 252	1	1.64
394 x 394 x 476	1	1.64
396 x 396 x 505	1	1.64
402 x 402 x 232.5	1	1.64
402 x 402 x 440	1	1.64
406 x 406 x 485	1	1.64
406 x 406 x 488	1	1.64
407 x 407 x 251.202	1	1.64
407 x 407 x 450	1	1.64
408 x 408 x 250	1	1.64
408 x 408 x 255	1	1.64
411 x 411 x 335	1	1.64
412 x 412 x 448	1	1.64
424 x 424 x 515	1	1.64
428 x 428 x 468	1	1.64
429 x 429 x 180	1	1.64
436 x 436 x 270	1	1.64
437 x 437 x 244	1	1.64
440 x 440 x 255	1	1.64
440 x 440 x 515	1	1.64
446 x 446 x 460	1	1.64
455 x 455 x 500	1	1.64

462 x 462 x 520	1	1.64
463 x 463 x 470	1	1.64
464 x 464 x 470	1	1.64
471 x 471 x 216.001	1	1.64
473 x 473 x 415	1	1.64
476 x 476 x 490	1	1.64
478 x 478 x 252	1	1.64
480 x 480 x 220	1	1.64
487 x 487 x 250	1	1.64
492 x 492 x 448	1	1.64
495 x 495 x 246	1	1.64
496 x 496 x 222	1	1.64
500 x 500 x 240	1	1.64
500 x 500 x 275	1	1.64
500 x 500 x 285	1	1.64
500 x 500 x 645	1	1.64

Dimensions XxYxZ | Patients | %

----- | ----- | ----

512 x 512 x 94	3	4.92
512 x 512 x 103	2	3.28
512 x 512 x 112	2	3.28
512 x 512 x 168	2	3.28
512 x 512 x 44	2	3.28
512 x 512 x 48	2	3.28
512 x 512 x 50	2	3.28
512 x 512 x 61	2	3.28
512 x 512 x 89	2	3.28
512 x 512 x 96	2	3.28
512 x 512 x 98	2	3.28
512 x 512 x 100	1	1.64
512 x 512 x 101	1	1.64
512 x 512 x 102	1	1.64
512 x 512 x 104	1	1.64
512 x 512 x 114	1	1.64
512 x 512 x 119	1	1.64
512 x 512 x 129	1	1.64
512 x 512 x 135	1	1.64
512 x 512 x 148	1	1.64
512 x 512 x 156	1	1.64
512 x 512 x 157	1	1.64
512 x 512 x 164	1	1.64
512 x 512 x 31	1	1.64
512 x 512 x 33	1	1.64
512 x 512 x 34	1	1.64
512 x 512 x 36	1	1.64
512 x 512 x 38	1	1.64
512 x 512 x 40	1	1.64
512 x 512 x 41	1	1.64
512 x 512 x 51	1	1.64
512 x 512 x 54	1	1.64
512 x 512 x 55	1	1.64
512 x 512 x 56	1	1.64
512 x 512 x 60	1	1.64
512 x 512 x 65	1	1.64
512 x 512 x 67	1	1.64
512 x 512 x 77	1	1.64
512 x 512 x 80	1	1.64
512 x 512 x 83	1	1.64

512 x 512 x 86	1	1.64
512 x 512 x 87	1	1.64
512 x 512 x 88	1	1.64
512 x 512 x 90	1	1.64
512 x 512 x 92	1	1.64
512 x 512 x 93	1	1.64
512 x 512 x 95	1	1.64
512 x 512 x 97	1	1.64
512 x 512 x 99	1	1.64

Intensity

=====

Statistic	Unit	Min	Max	Mean	Median
Minimum intensity	HU	-1024	-1024	-1024	-1024
Maximum intensity	HU	1166	4154	2003.836	1616
Mean intensity	HU	-759.276	-339.287	-534.88	-529.301
Standard deviation	HU	435.926	522.35	486.576	487.92
Percentile	Min	Max	Mean	Median	
P0.5	-1024	-1024	-1024	-1024	
P1	-1024	-1024	-1024	-1024	
P5	-1024	-1002	-1022.033	-1024	
P25	-1024	-942	-1000.607	-1001	
P50	-998	-110	-676.492	-843	
P75	-844	31	-57.59	-46	
P95	65	191	119.164	119	
P99	178	521	316.738	313	
P99.5	263	720	447.426	455	
Total voxels: 1,383,333,888 voxels					
Valid voxels: 1,383,333,888 voxels					
Non-finite voxels: 0 voxels					

Warnings

=====

20 patient(s) have missing data.
Voxel spacing varies across CT images.
Slice thickness varies across CT images.
Volume dimensions are not consistent across CT images.

Preprocessing Recommendations

=====

Severity	Category	Recommendation
high	annotations	Check missing labels or separate .
high	spatial_resampling	Resample CT images and labels to .
medium	slice_thickness	Standardize slice thickness or us.
medium	shape_standardization	Crop, pad, resize, or patch volum.
medium	intensity_normalization	Apply CT intensity clipping and n.
info	memory	Plan at least the estimated minim.